1. Current Business Situation

AtliQ Hardware sells computer hardware and peripherals to customers across multiple regions — India, North America, Latin America, Europe, and parts of APAC. The business operates through three main channels: brick-and-mortar retail, e-commerce platforms, and direct sales. Products are grouped into divisions — PC, Peripherals & Accessories, and Networking & Storage — each with its own revenue and margin profile.

Until recently, all reporting was done in Microsoft Excel. Individual teams maintained their own workbooks, pulling data from the source systems and building reports manually each cycle. For a business of moderate size, this was manageable. As AtliQ expanded into new geographies and added more product lines, the cracks started to show.

By the time the decision to move to Power BI was made, the analytics team was spending the majority of its time producing reports rather than analysing them. Leadership was reviewing numbers that were days old. Departments were working from different versions of the same metrics. And a failed market expansion — where strategic decisions were made without reliable data — had already resulted in tangible financial loss.

Context	*The Latin American market entry is cited as a key trigger. Decisions were made based on incomplete Excel analysis, and the outcome underscored the cost of operating without a proper analytics infrastructure.*

2. Existing Challenges

The challenges were not isolated to one team — they ran across every department that depended on data to operate.

Department	Challenge	Business Impact
Finance	No live P&L view. Gross Margin and Net Profit had to be assembled manually from multiple Excel files each month.	Delayed financial reviews; targets tracked retrospectively rather than in-cycle.
Sales	No single source for customer or product performance. Discount impact on margins was not visible without custom analysis.	Sales strategies built on incomplete profitability data; high-performing and loss-making accounts treated the same.
Marketing	Operational expenses tracked separately from revenue. No way to see how spend was affecting Gross Margin or Net Profit in real time.	Marketing decisions disconnected from financial outcomes; spend vs return difficult to justify.

Supply Chain	Forecast accuracy calculated manually and inconsistently. No standard way to flag overstock or stockout risk across products.	Reactive inventory management; excess stock and stockouts discovered after the fact.
Leadership	No consolidated executive view. Revenue, market share, and top-performer data came from different reports with different cut-off dates.	Strategic decisions made on fragmented, time-lagged information.
Analytics Team	Most time spent on report production — data pulls, formatting, distribution — rather than analysis or insight generation.	Low analytical output per cycle; team capacity wasted on manual work.

Beyond the department-level issues, there were systemic problems with the data itself. With no centralised data model, the same metric could be defined differently across teams. Net Sales in the finance workbook did not always match the figure used by sales. Gross Margin calculations varied depending on whether pre-invoice or post-invoice discounts had been factored in. Reconciling these differences took time and eroded trust in the numbers.

3. Why This Project Was Needed

The business had reached a point where continuing with Excel was not a resourcing problem — it was a structural one. No amount of additional effort on spreadsheets would resolve the underlying issues: inconsistent definitions, manual refresh cycles, no drill-down capability, and no way for multiple users to access current data simultaneously.

Three factors made this project urgent rather than optional:

- The Latin America market entry had already demonstrated that decisions made without reliable data had real financial consequences. The business could not afford a repeat.
- Global expansion was continuing. New markets meant more data, more complexity, and more reporting demand — all of which made the Excel model increasingly unworkable.
- Competitor intelligence suggested peers were investing in BI capabilities. Falling behind on analytics infrastructure had a strategic risk dimension beyond just operational efficiency.

Top management formally decided to adopt Power BI as the company's analytics platform, with the analytics team tasked to deliver a fully functional, department-specific dashboard — alongside SQL-based reporting for product owners who needed structured query access.

4. Proposed Solution

The solution was designed in two parts, addressing both the immediate reporting need and the longer-term need for self-serve analytics.

Part 1 — SQL Layer

The source data was loaded into MySQL and structured using a Snowflake schema. SQL queries were written to produce key reports: P&L metrics, top and bottom products and customers by net sales, and supply chain forecast accuracy. Stored procedures were created for the more complex queries so product owners could extract reports with filters — by time period, market, or product segment — without needing to write SQL themselves.

Part 2 — Power BI Dashboard

Power BI connects directly to MySQL and Excel via native connectors. Data is transformed in Power Query — handling nulls, standardising column names, merging tables, and deriving fields before they enter the model. The data model follows the same Snowflake schema, and all KPIs are built as DAX measures to ensure consistency across all views.

The dashboard has five pages, each designed around the needs of a specific audience:

- Finance View — P&L statement from Gross Sales to Net Profit, with benchmark comparisons against last year or target.
- Sales View — Customer and product performance with unit economics: discount breakdowns, COGS, and Gross Margin at account level.
- Marketing View — Operational expense impact on Net Profit, with a performance matrix across segments and categories.
- Supply Chain View — Forecast Accuracy %, Net Error, and ABS Error with risk flags and trend analysis by customer and product.
- Executive View — Market share trend, revenue by channel and division, top 5 products and customers, and sub-zone contribution analysis.

The report was published to Power BI Service, with a data gateway enabling automatic refresh from MySQL and Excel. Stakeholder feedback gathered during UAT led to the addition of the Executive View and several quality fixes before the final deployment.

5. Expected Business Impact

The shift from Excel to a governed BI environment change how the business operates at a practical level:

Area	Before	After
Reporting Cycle	Days to produce; manual, error-prone	Automated refresh; current data always available
Metric Consistency	Definitions varied by team and workbook	Single data model; one definition per KPI
Finance Visibility	Monthly P&L assembled manually	Live P&L with target vs actuals at any level
Sales Intelligence	No customer-level margin visibility	Gross Margin and discount impact per account
Supply Chain Risk	Stockouts and overstock caught reactively	Forecast accuracy and risk flags visible in advance
Executive Reporting	Multiple disconnected reports, different dates	Single Executive View with all key metrics
Analyst Productivity	Majority of time on report production	Freed for actual analysis and stakeholder support

The immediate measurable outcome is a reduction in manual reporting effort and faster access to current KPIs. The longer-term outcome — better decisions, faster course correction, and fewer instances of strategy built on bad data — is harder to quantify but was the actual driver behind the project.